SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Pattern Recognition and Text Processing

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 1 – Pattern Recognition and Text Processing
Weightage:	40% of CIA		
CO5 Statement:	Analyse regular expression patterns. (BL-3)		
Student Name:	T. Jyothi Reddy	**Reg. No.:**	192472079
Section / Batch:	CSE(AI)	**Date:**	

Code of Conduct

I T. Jyothi Reddy (192472079) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: T.J TR

Instructions

- Show all intermediate steps, calculations, state transitions, and reasoning wherever applicable.
- Use only the Python re (Regular Expressions) module to solve the given problems.
- Clearly mention the Regular Expression (Regex) pattern used before writing the corresponding Python program.
- Display the required output for each question, including extracted values, validation results, counts, and positions wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Generation	Pattern Matching Information Extraction	1
Search for a String	Keyword Search and Text Analysis	2
Split the documentation into sentences and words	Text Tokenization and Sentence Segmentation	3
Email and Strong Password Validation	Data Validation and Format Verification	4
Compile Once, Validate Many	Pattern Reusability and Performance Optimization	5

100

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q. No. / Criteria	Conceptual Understanding	Accuracy of Answer	Application of Knowledge	Completeness of Response	Clarity & Presentation	Sub. Total	Total %
1						20	
2						20	
3	4		4	4	4	20	
4		4				20	100
5						20	

Faculty In-charge	Course Coordinator	Program Director
Dr. T. Kumaraguruparan.		
Signature: ____________	Signature: ____________	Signature: ____________
Date: 14/7/26.	Date: ____________	Date: ____________

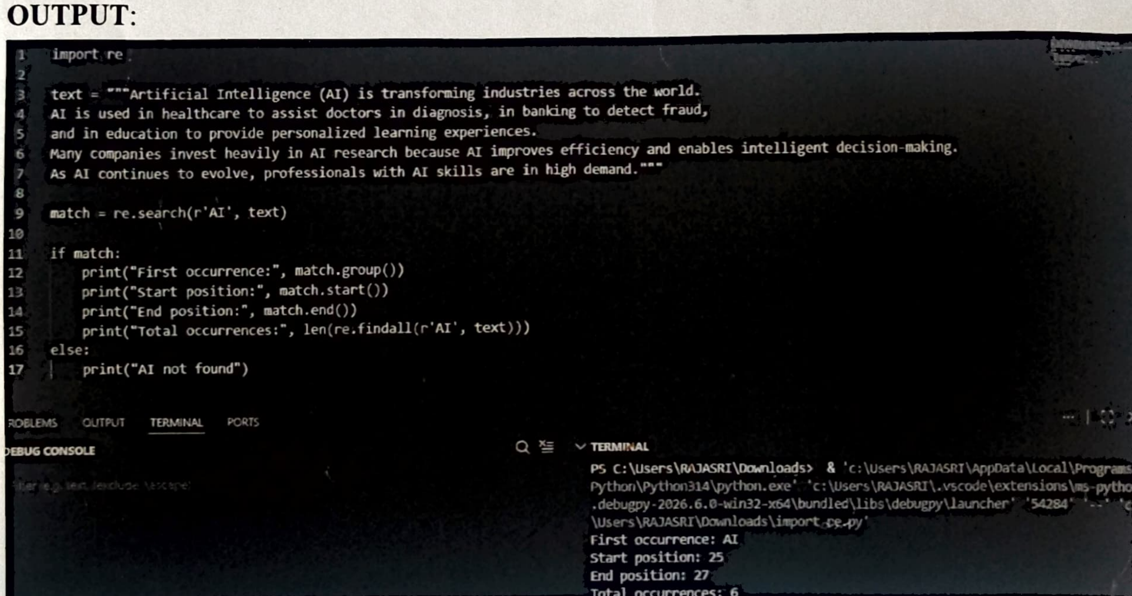
Q1. Regular Expression Pattern Generation

Email = r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$'
Mobile = r'^\+91-\d{10}$'
Password = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#$%&!*])[A-Za-z\d@#$%&!*]{8,}$'
DOB = r'^\d{2}/\d{2}/\d{4}$'
Register = r'^\d{2}[A-Z]{5}\d{4}$'

Q2. Search for a String

```
import re
text = """Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare. Many companies invest heavily in AI research because AI improves efficiency. As AI continues to evolve, professionals with AI skills are in high demand."""
match = re.search(r'AI', text)
if match:
    print(match.group())
    print(match.start())
    print(match.end())
    print(len(re.findall(r'AI', text)))
else:
    print("AI not found")
```

OUTPUT:



```
import re

text = """Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud,
and in education to provide personalized learning experiences.
Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making.
As AI continues to evolve, professionals with AI skills are in high demand."""

match = re.search(r'AI', text)

if match:
    print("First occurrence:", match.group())
    print("Start position:", match.start())
    print("End position:", match.end())
    print("Total occurrences:", len(re.findall(r'AI', text)))
else:
    print("AI not found")
```

ROBLEMS OUTPUT TERMINAL PORTS

DEBUG CONSOLE

TERMINAL

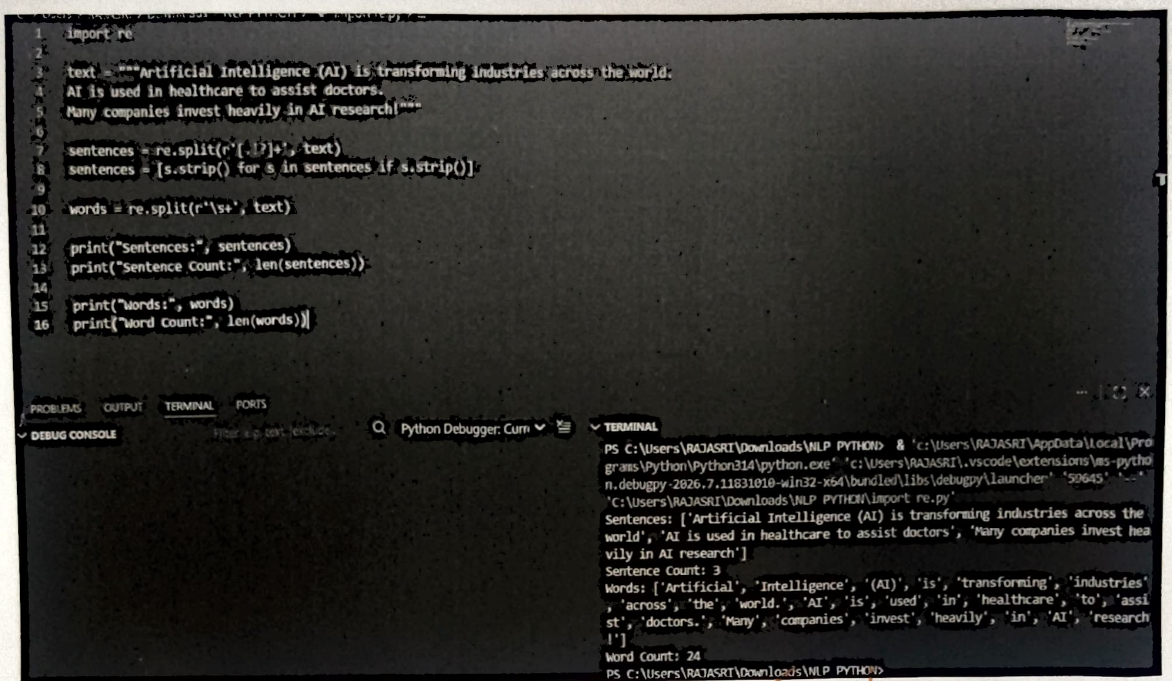
```
PS C:\Users\RAJASRI\Downloads> & 'c:\Users\RAJASRI\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\RAJASRI\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '54284' '--' 'c:\Users\RAJASRI\Downloads\import re.py'
First occurrence: AI
Start position: 25
End position: 27
Total occurrences: 6
```

Q3. Split into Sentences and Words

```
import re
sentences=re.split(r'[.!?]+',text)
sentences=[s.strip() for s in sentences if s.strip()]
words=re.split(r'\s+',text)
print(sentences)
print(len(sentences))
```

print(words)
print(len(words))

OUTPUT :



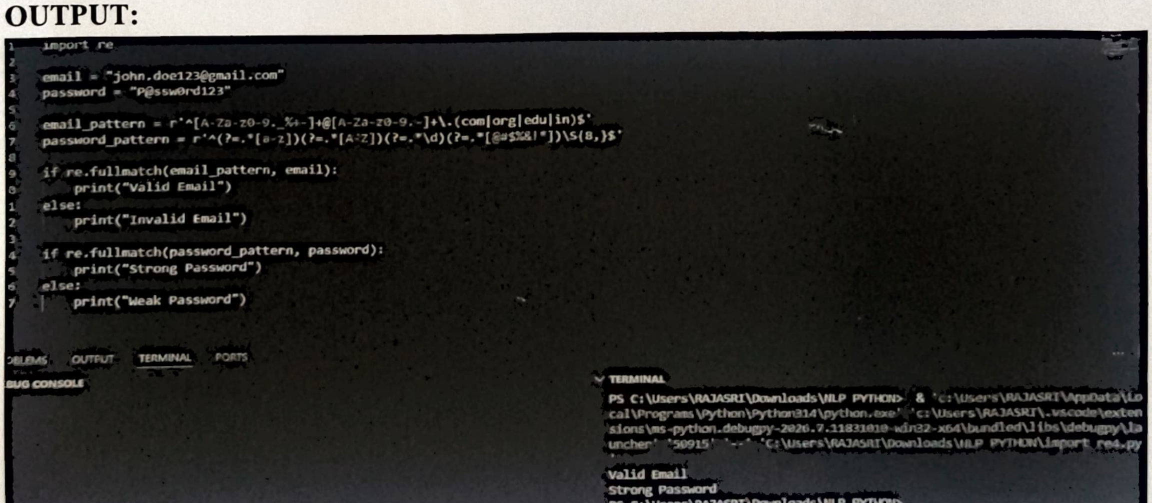
```
import re

text = """Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors.
Many companies invest heavily in AI research!"""

sentences = re.split(r'[.!?]+', text)
sentences = [s.strip() for s in sentences if s.strip()]

words = re.split(r'\s+', text)

print("Sentences:", sentences)
print("Sentence Count:", len(sentences))

print("Words:", words)
print("Word Count:", len(words))
```

```
PS C:\Users\RAJASRI\Downloads\NLP PYTHON> & 'c:\Users\RAJASRI\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\RAJASRI\.vscode\extensions\ms-python.debugpy-2026.7.11831010-win32-x64\bundled\libs\debugpy\launcher' '59645' '--' 'C:\Users\RAJASRI\Downloads\NLP PYTHON\import re.py'
Sentences: ['Artificial Intelligence (AI) is transforming industries across the world', 'AI is used in healthcare to assist doctors', 'Many companies invest heavily in AI research']
Sentence Count: 3
Words: ['Artificial', 'Intelligence', '(AI)', 'is', 'transforming', 'industries', 'across', 'the', 'world.', 'AI', 'is', 'used', 'in', 'healthcare', 'to', 'assist', 'doctors.', 'Many', 'companies', 'invest', 'heavily', 'in', 'AI', 'research!']
Word Count: 24
PS C:\Users\RAJASRI\Downloads\NLP PYTHON>
```

Q4. Email and Password Validation

import re
email_pattern=r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$'
password_pattern=r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#$%&!*])\S{8,}$'
print("Valid Email" if re.fullmatch(email_pattern,"john.doe123@gmail.com") else "Invalid Email")
print("Strong Password" if re.fullmatch(password_pattern,"P@ssw0rd123") else "Weak Password")

OUTPUT:



```
import re

email = "john.doe123@gmail.com"
password = "P@ssw0rd123"

email_pattern = r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$'
password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#$%&!*])\S{8,}$'

if re.fullmatch(email_pattern, email):
    print("Valid Email")
else:
    print("Invalid Email")

if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")
```

```
PS C:\Users\RAJASRI\Downloads\NLP PYTHON> & 'c:\Users\RAJASRI\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\RAJASRI\.vscode\extensions\ms-python.debugpy-2026.7.11831010-win32-x64\bundled\libs\debugpy\launcher' '50915' '--' 'C:\Users\RAJASRI\Downloads\NLP PYTHON\import re4.py'
Valid Email
Strong Password
PS C:\Users\RAJASRI\Downloads\NLP PYTHON>
```

Q5. Compile Once, Validate Many

```
import re
emails=["john@gmail.com","abc@yahoo.org","test@college.edu","wrong@email","user123@c
ompany.in"]
pattern=re.compile(r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$')
for e in emails:
    print(e,"Valid" if pattern.fullmatch(e) else "Invalid")
```

OUTPUT :